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SOIL AND WATER CONSERVATION IN TIGRAY (NORTHERN ETHIOPIA): THE TRADITIONAL DAGET TECHNIQUE AND ITS INTEGRATION WITH INTRODUCED TECHNIQUES

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ABSTRACT

Between the cultivated fields in the highlands of Tigray, one finds, besides the recently introduced stone bunds, many lynchets, with a height ranging from 0·3–3 m. Grasses occupy the riser and a more or less large strip on the shoulder.

Traditionally, farmers established an untilled strip of about 2 m wide at the lower plot limit. This grass strip reduced runoff velocity, allowed for water to infiltrate and trapped sediment. Year after year, these lynchets, locally called daget, continued to grow. In this study, the daget technique is characterized and its evolution and reasons for partial destruction are outlined. Stereoscopic aerial photo analysis shows for the study area near Hagere Selam (13°39′N, 39°10′E, 2650 m a.s.l.) that 20·7 per cent of the major dagets (more than 1 metre high) have disappeared between 1974 and 1994. These lynchets, however, remain an important linear element in the landscape (22·7 m ha^{-1}, i.e. their density on cultivated fields is much higher). Of the smaller lynchets, a great proportion has been levelled in order to increase plot surfaces and spread fertile soil over the field. Famines and impoverishment caused the farmers to increase short-term agricultural production in this way.

Since the 1980s, the farmers built stone bunds on most of the cultivated land. Their average length equals 56·1 m ha^{-1} in the study area. The establishment of stone bunds results in the development of small terraces. Especially during recent years, there is a tendency to integrate the traditional knowledge of daget with the building of stone bunds.

A quantitative assessment of the effectiveness of both techniques must be made.

KEY WORDS: soil conservation techniques; water conservation techniques; lynchet; grass strip; stone bund; highlands; Ethiopia; Tigray

INTRODUCTION

The agricultural system in the northern Ethiopian highlands has been rightly qualified as 'grain-plough complex' (Westphal, 1975). Barley, wheat and the endemic tef (*Eragrostis tef*) are most cultivated. The mahrasha, an implement with a wedge-shaped metal point, drawn by two oxen, which throws the soil onto both sides of the furrow, is used for ploughing.

In Ethiopia's northernmost region, Tigray, a land reform process has been completed (1978–90), granting land to the tiller.

Between the fields, one finds, besides the recently introduced stone bunds, many lynchets, with a height ranging from 0·3 m up to 3 m. Grasses occupy the riser and a more or less large strip on the shoulder. These lynchets, which are not necessarily on the contour, correspond mostly to pre-land-reform plot limits. In the local Tigrinya language, they are commonly called daget. The word armo, which was previously used (Nyssen, 1995; 1997), indicates any boundary, an alignment of stones, a strip of herbaceous vegetation or

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even the boundary of a building plot. The more restrictive word deret designates specifically the narrow strips of uncultivated land (Bruce, 1976: 462) but it is less used.

The daget are one of these 'elements of landscape which have not yet been characterized' (Showers, 1996). Reij (1991), in his general outline of traditional soil and water conservation techniques in Africa, briefly mentions these structures. Bono and Seiler (1986) analyse a soil sequence on an 'ancient, man-made terrace', which, according to the published sketch and picture is a lynchet. The best description is probably found in a study by the Ethiopian Ministry of Agriculture concerning traditional conservation techniques: (in Harar region in eastern Ethiopia) 'all farmers questioned claimed that, many years ago, strips of uncultivated land were left between fields. With time, the strips have trapped soil and terraces have been built up to their present height of 30–60 cm. Invariably, the terrace 'wall' was stabilized by bushes and trees (such as *Euphorbia* spp.) and sometimes, even by stone facing' (Ministry of Agriculture, 1988: C6). Krüger, *et al.* (1997) comment on the use of different shrub species for the reinforcement of 'bunds': the figure (Krüger, *et al.*, 1997: 55) shows an acacia line at the lower edge of a lynchet.

In Kenya, Kiepe and Young (1992) found that grass strips of 1·6 m wide, planted with trees (every 3 m) and alternating with 5-m-wide fields, involve, in a period of six years, the creation of about 0·4 high risers and a decrease of the field's slope from 13·8 to 6·8 per cent. This type of structure functions thus as a vegetative barrier (Sheng, 1989: 65): more precisely, a partial barrier for water-induced erosion (Lakew and Morgan, 1996) but a total barrier for tillage (dry mechanical) translocation (Quine, *et al.*, 1993; Turkelboom, *et al.*, 1997); the relative contribution of both erosion processes to the total amount of sediment deposited in the daget still needs to be further examined.

These lynchets are similar to those existing in Europe, where many of these also result from tillage erosion and water erosion on temporarily bare soils (in addition to excavation of brick earth) (Gerlach, 1963; Ozer, 1969; Bollinne, 1971; Van Oost and Govers, 1998).

We studied the daget in Dogua Tembien, a region some 40 km to the west of Tigray's regional capital Makalle (Figure 1) in an attempt to understand its genesis and its evolution, especially in relation to recent catchment terracing programmes (stone bund building).

STUDY METHODS

Fieldwork included topographical surveys of daget by laser theodolite and the drawing of cross-sections through these lynchets. Farmers were interviewed regarding present-day and previous locations and dimensions during fieldwork. This was preferred to the use of a standardized questionnaire because the sites are very variable (e.g. dimensions of the daget) and because there is no uniform terminology for the various (parts of) structures in the local Tigrinya language.

To assess the evolution of the soil conservation structures in the landscape, pairs of aerial photographs taken in 1974 (Hunting Technical Services, DOS Contract 140: ET 39 072 and 073; scale 1 : 25 000) and 1994 (Ethiopian Mapping Authority, ETH 94 R-2 25 753 and 754; scale 1 : 50 000, enlarged to 1 : 25 000) were interpreted by stereoscopy. The major daget (height estimated at 1 m or more during field visits) as well as the stone bunds are clearly discernible at that scale. The photo interpretations were digitized and compared after coregistration (ArcInfo® software).

RESULTS

The Daget: A Soil Conservation Technique

Several major daget, hundreds of metres long, can be observed on the southern slope of the Mehabere Selasse mountain, close to Hagere Selam. One of these (Figures 2 and 4) reaches in several places a height of 3 m and the slope of the riser varies between 53 per cent and 83 per cent.

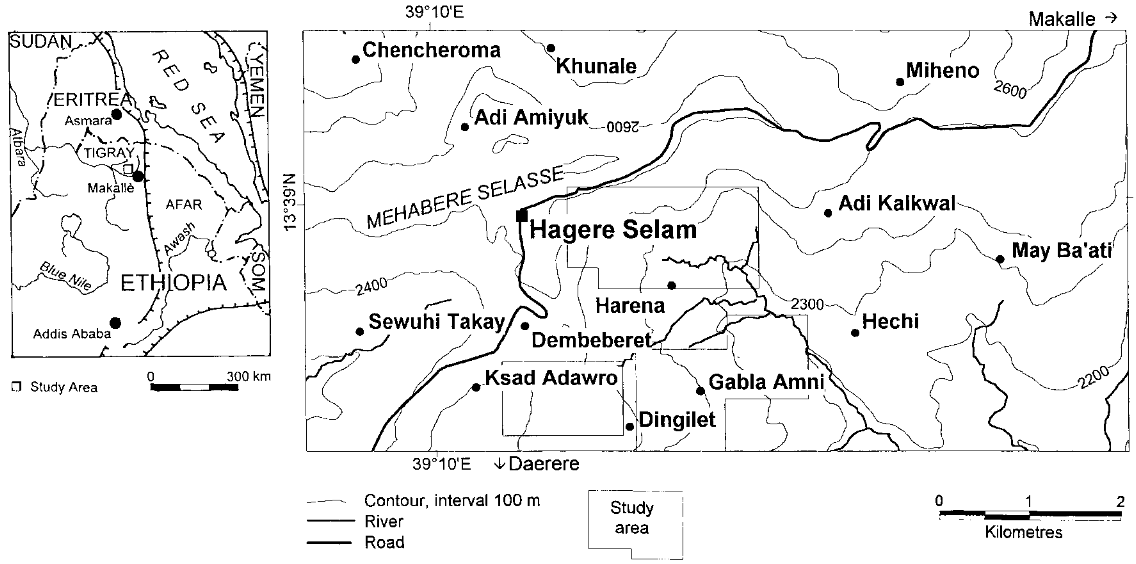

Figure 1. The study area in Dogua Tembien, central Tigray, northern Ethiopian highlands.

Figure 2. A major daget on the southern slope of Mehabere Selasse mountain, 700 m west of Hagere Selam (September 1997).

Figure 3. Lynchet with subvertical, retreating riser in Amba Gwasot, a hamlet of Daerere (September 1997).

In Daerere (Figure 3), the observed daget has a subvertical, retrograding riser 0·8 m high. It used to be higher with a less steep riser, but, according to the farmers, ploughing close to the riser induced failures, the material of which was deposited downslope, contributing to the raising of the lower field.

A typical smaller daget was measured in a field with a slope of 5 per cent in Hechi: its height varies between 0·44 and 0·66 m and the gradient of the riser between 46 per cent and 56 per cent. Many daget bear dense herbs. Most common grasses are *Cynodon dactylon* (t'hag) and *Pennisetum* sp. (tsada saeri). Locally, shrubs also reinforce the daget: e.g. *Acacia* div. sp., *Rumex nervosus* and *Nicotiana glauca*.

Older farmers spontaneously quote these daget as a technical implement to decrease soil erosion. The daget also marked off the parcels belonging to different farmers and supply pastureland and/or fodder for the cattle. These lynchets have been consciously maintained, generation after generation. At the lower side of the plot, an often 2 m wide grass strip (interviewed farmers indicated this previous width by making two or three steps) was never ploughed: runoff velocity was reduced by the grasses, which facilitated infiltration and deposition of sediments.

A cross-section through a daget on the slope of Mehabere Selasse mountain (Figure 4) allows a better understanding of the process. After deforestation, probably some 2500 years ago (Hurni, 1985), topsoil and subsoil were removed, predominantly by water and tillage erosion. The weathered parent material – a silicified lacustrine deposit, interbedded between Tertiary trapp basalt layers (Arkin, *et al*., 1971) – appeared on the surface of the field. The clear discontinuity between this truncated profile (without traces of pedogenesis), and the overlying materials, as well as the presence of historical plough furrows at the top of the weathered parent material (Figure 5) are witnesses of this dramatic palaeomorphological process. Since the establishment of the grass strip (armo), the accumulation of sediment resulted in a daget, which overlies the level of the previous field by approximately 2 m. This is attested by the presence of a tilled horizon in the cross-section, in the lower part of the overlying colluvium. The present field is almost horizontal on the shoulder of the daget; this must be compared to the original slope of some 40 per cent at the moment of the establishment of the grass strip. Slumping occurs in the riser. This is caused by close ploughing at its foot, which increases the slope of the riser to a gradient greater than the angle of rest of the accumulations. Before slumping, the daget might have shown a profile similar to the one indicated on Figure 4.

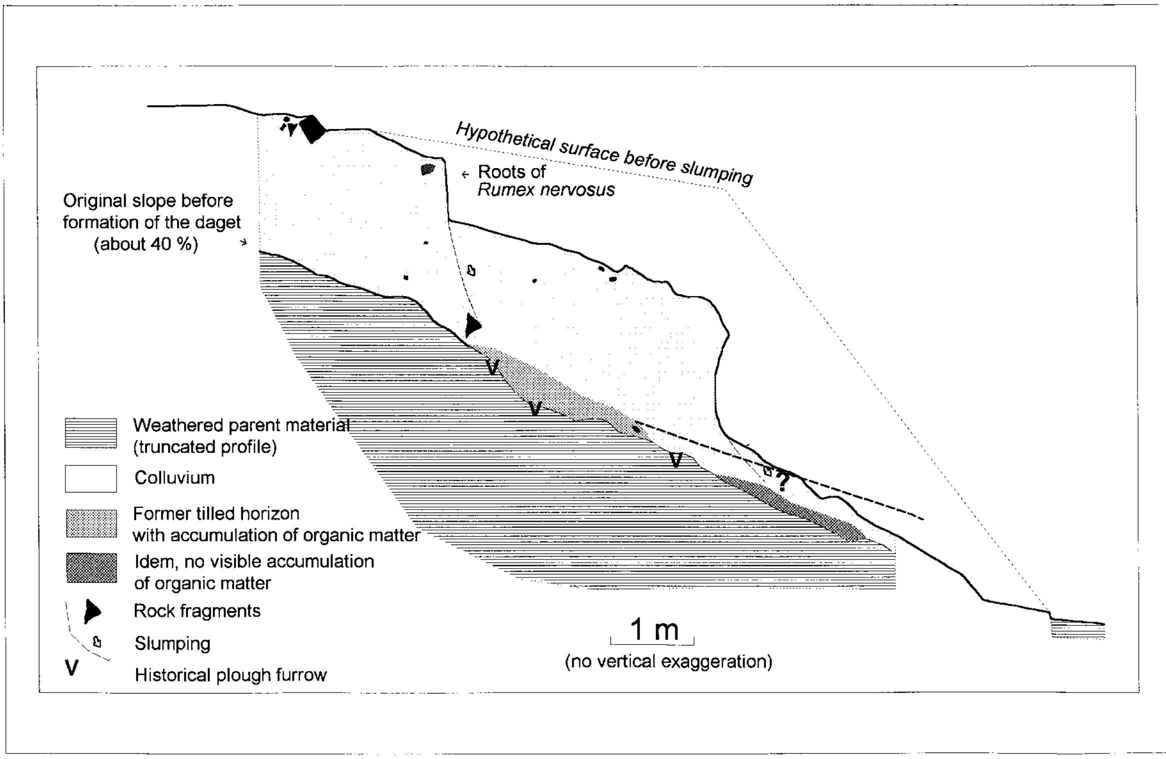

Figure 4. Cross-section through a daget, close to Hagere Selam (see also Figure 2).

Figure 5. The presence of historical plough furrows (arrows) shows the importance of colluviation in the daget.

Evolution of the Daget since 1974

Some farmers claim that many daget have been destroyed over the last decades; a temporal relationship is sometimes made with land reforms initiated after the overthrowing of the Emperor in 1974 and concomitant changes in the agrarian morphology. Before, often, 2 m wide grass strips (armos) existed at the bottom of the plots, and 'there was no need to make footpaths through the crops as people do nowadays. We could walk on armos everywhere!'

To assess the evolution of the landscape, aerial photographs were analysed by stereoscopy. In the analysed area of 543·3 ha including arable and grazing land (see Figure 1), the clearly visible daget (during field visits, their height was estimated to be more than 1 m) totalled 15 558 m in 1974. Out of these, 3224 m (20·7 per cent) had disappeared in 1994. The lynchets are still clearly present in the landscape in Dogua Tembien (22·7 m of major daget per hectare).

Rather than these major landscape elements, changes mainly concern the reduction of the width of border grass strips as well as the destruction of numerous daget with an elevation of less than 1 m.

We observed three activities that contribute to the destruction of the daget: the owner of the downslope plot may plough very close to its foot, which results in frequent subvertical and retreating risers due to mass movement processes. Furthermore, like most vegetation in between the fields, the width of the armo (grass strip) on the shoulder has generally been reduced, and this results in increased runoff (less sediment trapping and more water induced erosion). Sometimes the (smaller) daget are deliberately destroyed by hoe (Figure 6).

DISCUSSION

Reasons for Destruction: Poverty and Social Insecurity

In Dogua Tembien, even now with an increasing agricultural production, average yields are sufficient for feeding the population during 10 months per year only; the peasants are thus dependent on food aid and other survival strategies and/or suffer from malnutrition and subsequent illnesses. Infant mortality is high. These impoverished farmers (for a historical explanation of this poverty, see Ståhl, 1974; Girma and Jacob,

Figure 6. A smaller daget, destroyed by hoe (Zenak'o, between Hagere Selam and Adi Kalkwal, April 1998). Before destruction, it was similar to the small lynchets situated behind it.

1988; Nyssen, 1995) have lived through two great famines (1972–73 and 1984–85) during the last 25 years. Even now, they give examples showing that the slightest decrease in agricultural production can be fatal for one of the members of the family and they try to increase the surface suitable for cultivation. Especially since the 1984–85 famine, many smaller daget have been destroyed to increase the field's size and to recover the accumulated soil rich in organic matter.

In short, in situations of poverty and social insecurity, short-term survival will prevail over medium- and long-term conservation issues.

Integration of the Traditional Daget Technique with the Introduced Stone Bunds

The daget are only present in part of the fields and are generally discontinuous. Their present density is not sufficient to control soil erosion in whole catchments. Accordingly, since the 1980s, the stone bund technique, which existed already in a scattered way on the Ethiopian highlands (Buxton, 1949; Huffnagel, 1961), has been popularized in Tigray (REST, 1984; Robinson, 1987; Nyssen, 1998). Like almost everywhere in this region, the larger part of the arable land in Dogua Tembien has been terraced by community work (Table I). In the total analysed area, 30 498 m (overall density: 56·1 m ha^{-1}) of new stone bunds appeared on the contour, which induced sediment deposition resulting in terraces between 0·4 and 1·2 m high.

Stone bunds have the advantage, as compared with daget, of using less of the field's surface, of being level and of being laid out systematically in large parts of watersheds. Kloosterboer and Eppink (1989: 132) show, in the Cape Verde Islands, that risks of induced gullying are less when slopes are treated in an integrated way. These physical structures are sometimes on the shoulder of a subsisting daget, raising it artificially. In other areas, bunds are not built on the pre-existing daget, particularly if these are not on the contour or if their spacing does not correspond to the terraces' spacing (Figure 7). But sometimes, stone bunds are situated only a few metres up- or downslope of a daget and parallel to it, or cross it at a small angle. Hence, the farmer will be tempted to destroy the daget (and not the stone bund that has been built recently by community labour). Asked why such a stone bund was not built on the nearby daget, without thus taking advantage of an existing terrace, one extension agent answered: 'The instrument (a line-level) told us to build it here.' At the onset, the dense stone bund network was partially superimposed on (i.e. not integrated with) the traditional soil conservation structures. At present, one more often sees that the layout of the stone bunds includes the existing daget, as suggested by some handbooks and guidelines (Hurni, 1986; BNREP and REST, 1994).

The regional Bureau of Agriculture also advises leaving a 0·5 m wide strip unploughed along the stone bunds, on the upper side. Like an armo, the resulting vegetation strip will decrease runoff velocity and consolidate the stone bund. These instructions, which correspond to the ancestral armo technique, are followed quite well in the lower areas, which are situated on limestone: in Hechi or Daerere one sees often

Table I. Recommended spacing for stone bunds in Tigray (BoANR, 1997)

Slope (%)	Horizontal interval (m)
5	20
6	17
7–8	14
9–11	13
12–15	12
16–17	11
18–19	10
20–23	9
24–25	8
26–29	7
30–34	6
⩾35	5

Figure 7. A flight of five daget (not on contour), along the road to Makalle, 10 km east of Hagere Selam (January 1998). A pattern of parallel recent stone bunds on the contour is superimposed on it. On two locations (arrows), the farmers found it, however, more economical to build the stone bunds on top of the existing daget.

grass strips up to 1 m wide along the stone bunds. This integration of techniques, however, is less used on the basalt-derived clayey soils in the highlands. In less productive areas (the lower areas in Dogua Tembien have not only poor and shallow soils but also less rainfall and more evapotranspiration), farmers are willing not to plough some land for the sake of soil and water conservation. But generally, the farmers do not give up strips of the so badly needed productive land for the establishment of grass strips.

Better access to facilities (fertilizer, improved seeds, tools, ...), as well as successful collaboration between peasant farmers and technicians that are aimed at (Aseffa, *et al.*, 1992; Deckers, 1998), lead, however, to an increased agricultural output per cultivated surface unit. For some years, Ethiopia has yielded bumper harvests. (In 1997, a drought occurred comparable to the one in 1984, but famine could not be avoided.) Such intensification leaves the possibility for the maintenance and improvement of landscape elements like daget, as was also observed in the Machakos district in Kenya (Tiffen, *et al.*, 1994).

CONCLUSIONS

The daget are made up of colluvium: on its way downslope, soil translocated by tillage and water erosion is trapped in the grass strip and in the backwater that the grass strip creates during important runoff events.

Most of the major daget in Dogua Tembien have not been destroyed since 1974 and continue to contribute to soil and water conservation, but minor ones and grass strips have often been removed in a bid to increase the surface of agricultural land. The integration of traditional and well-thought-out, modern soil and water conservation techniques is possible: in Tigray, the introduced stone bunds often rise on the existing daget. Moreover, especially in the less productive areas, it can also be observed that the farmers establish grass strips along the stone bunds, in the same way as they were accustomed to do with the daget.

However, a quantitative assessment of the daget's effectiveness as an erosion control technique and a comparison with stone bunds still needs to be made.

In view of the deliberate destruction of daget, as well as the ploughing that is carried out close to the upper and lower side of the riser, it seems clear that the awareness of the authorities and the extension agents for the protection of daget must be increased. Likewise, when popularizing soil and water conservation techniques, the inclusion of daget and other terrain irregularities in the stone bund network should be stressed.

Finally, the intensification of agriculture and the increase of the output per cultivated surface unit could alleviate the pressure on these landscape units, which are important for soil and water conservation. This requires increased investment in agricultural development.

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